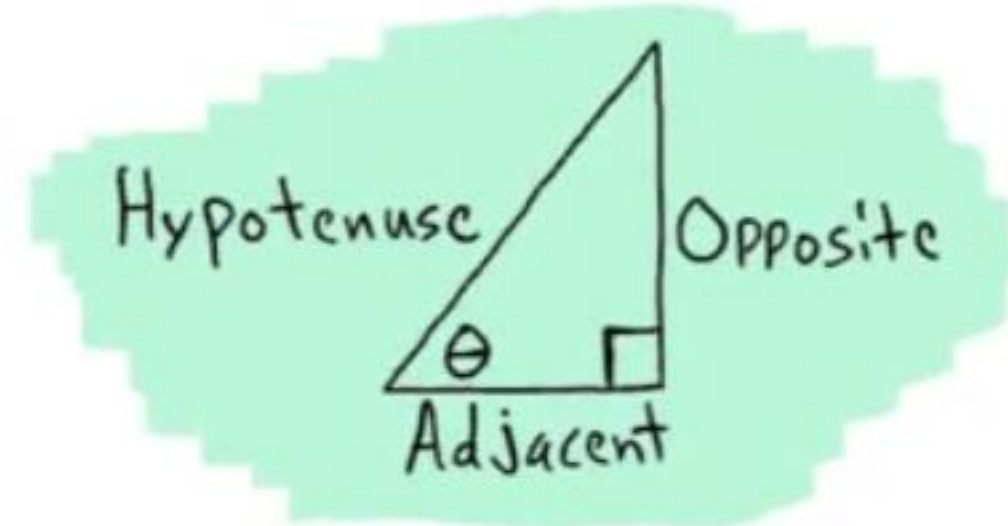


Memorize These



$$\sin \theta = \frac{\text{OPP}}{\text{HYP}}$$

$$\cos \theta = \frac{\text{Adj}}{\text{HYP}}$$

$$\tan \theta = \frac{\text{OPP}}{\text{Adj}}$$

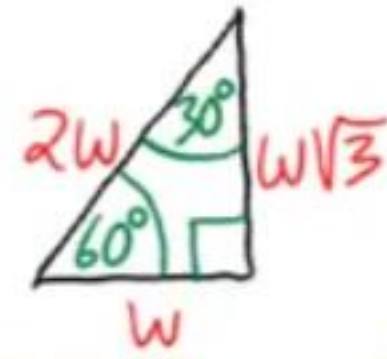
$$\csc \theta = \frac{\text{HYP}}{\text{OPP}}$$

$$\sec \theta = \frac{\text{HYP}}{\text{Adj}}$$

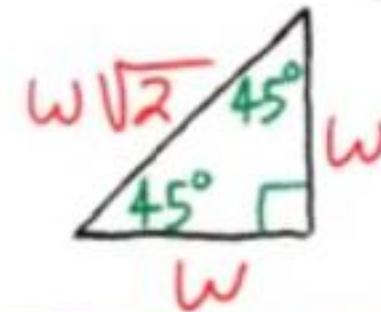
$$\cot \theta = \frac{\text{Adj}}{\text{OPP}}$$

Special Triangles

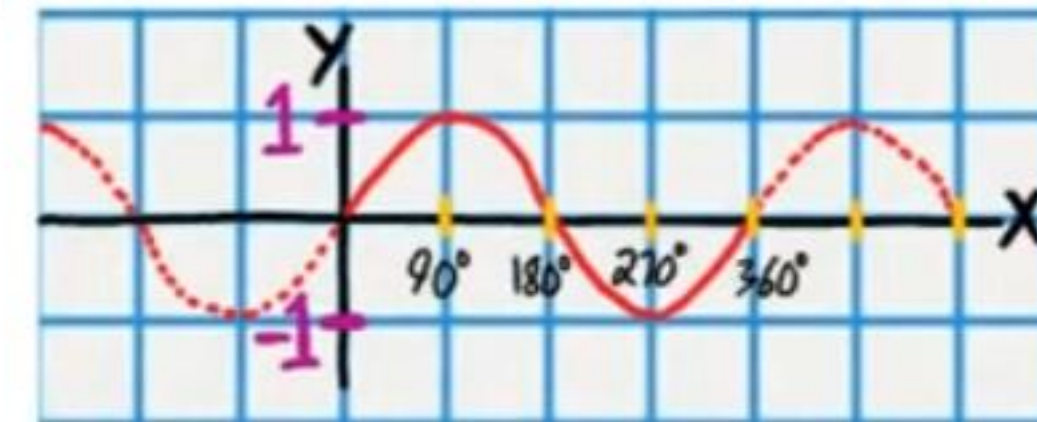
30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60° and 90°, and the length of the side opposite the 30° angle is w , then the length of the hypotenuse is $2w$ and the length of the side opposite the 60° angle is $w\sqrt{3}$.



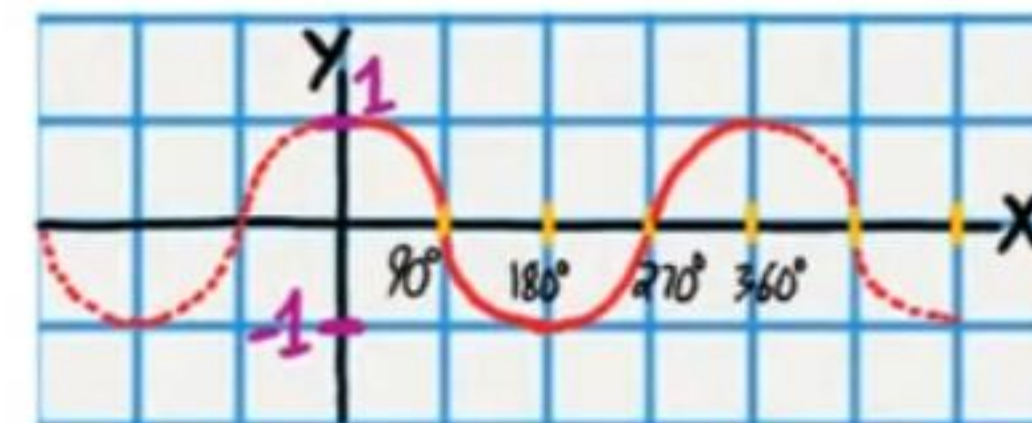
45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45° and 90°, and the length of the two sides opposite the 45° angles are w , then the length of the hypotenuse is $w\sqrt{2}$.



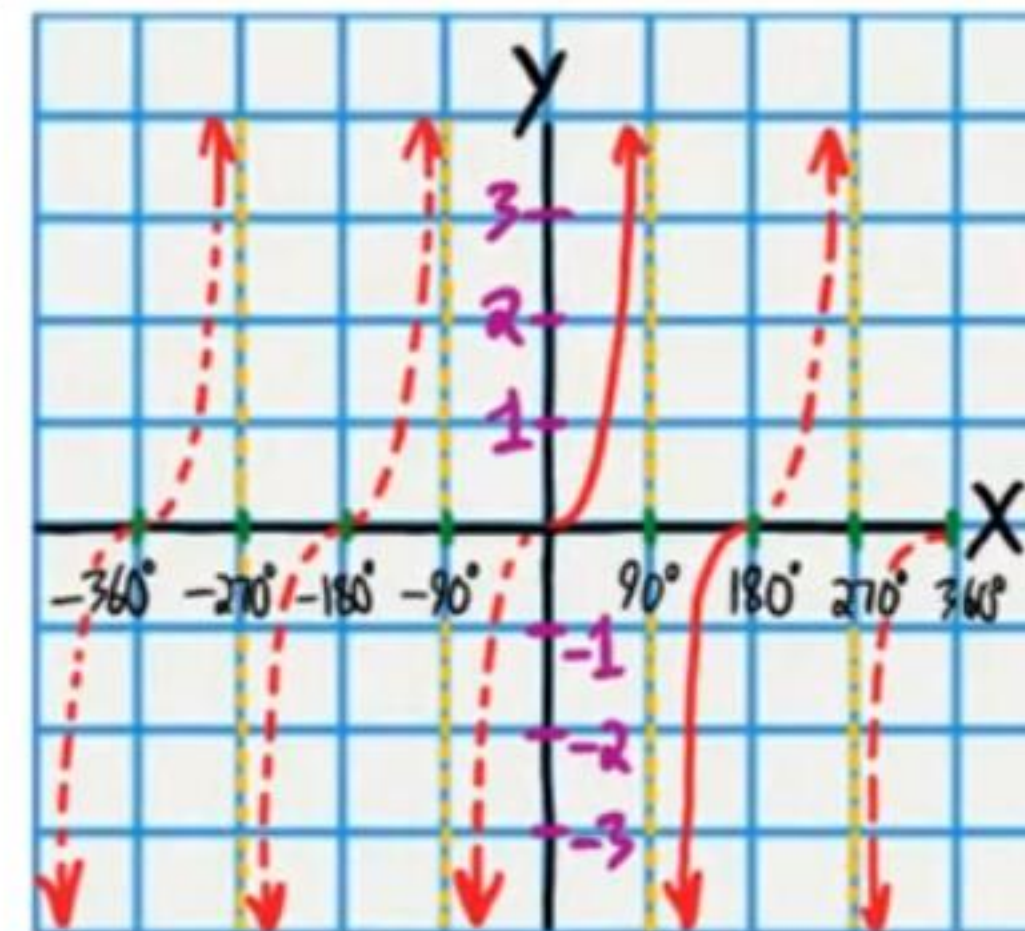
$$y = \sin x$$



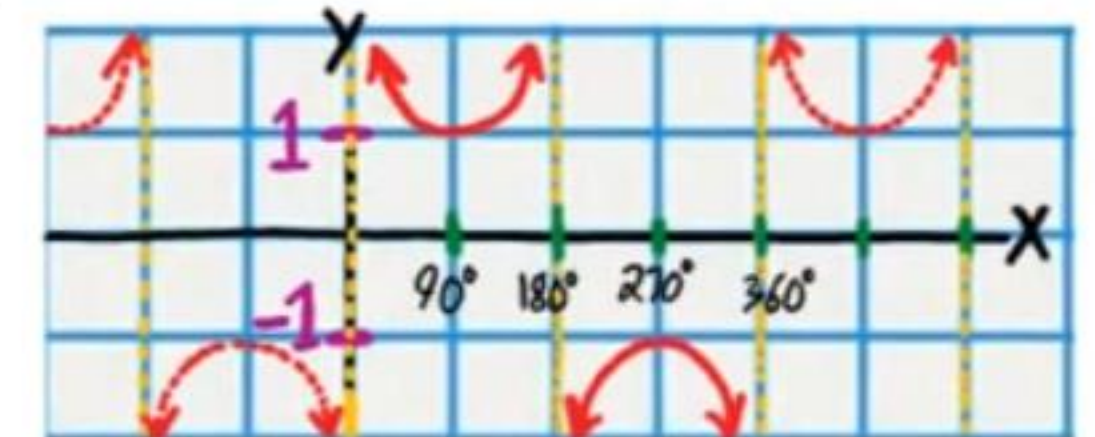
$$y = \cos x$$



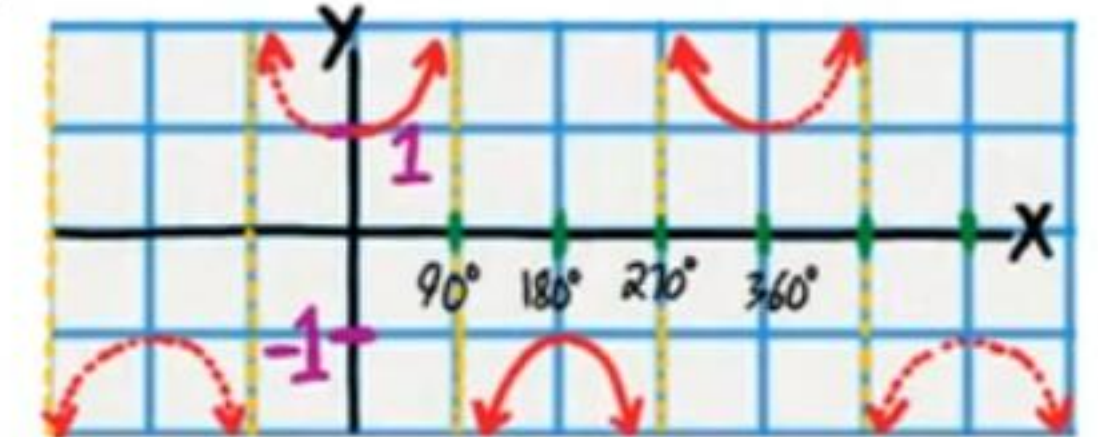
$$y = \tan x$$



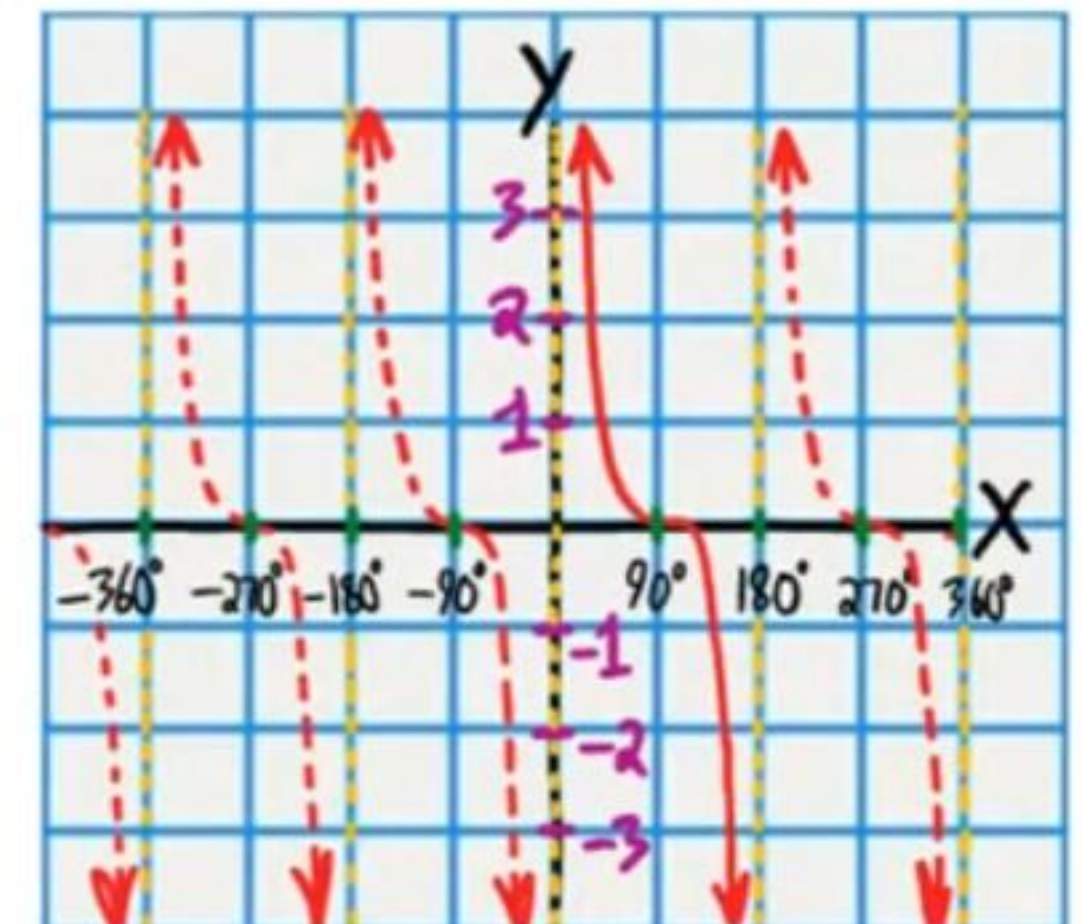
$$y = \csc x$$



$$y = \sec x$$



$$y = \cot x$$



Practice Test I (Version 1)

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Trigonometry).

II) Show all work.

III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.

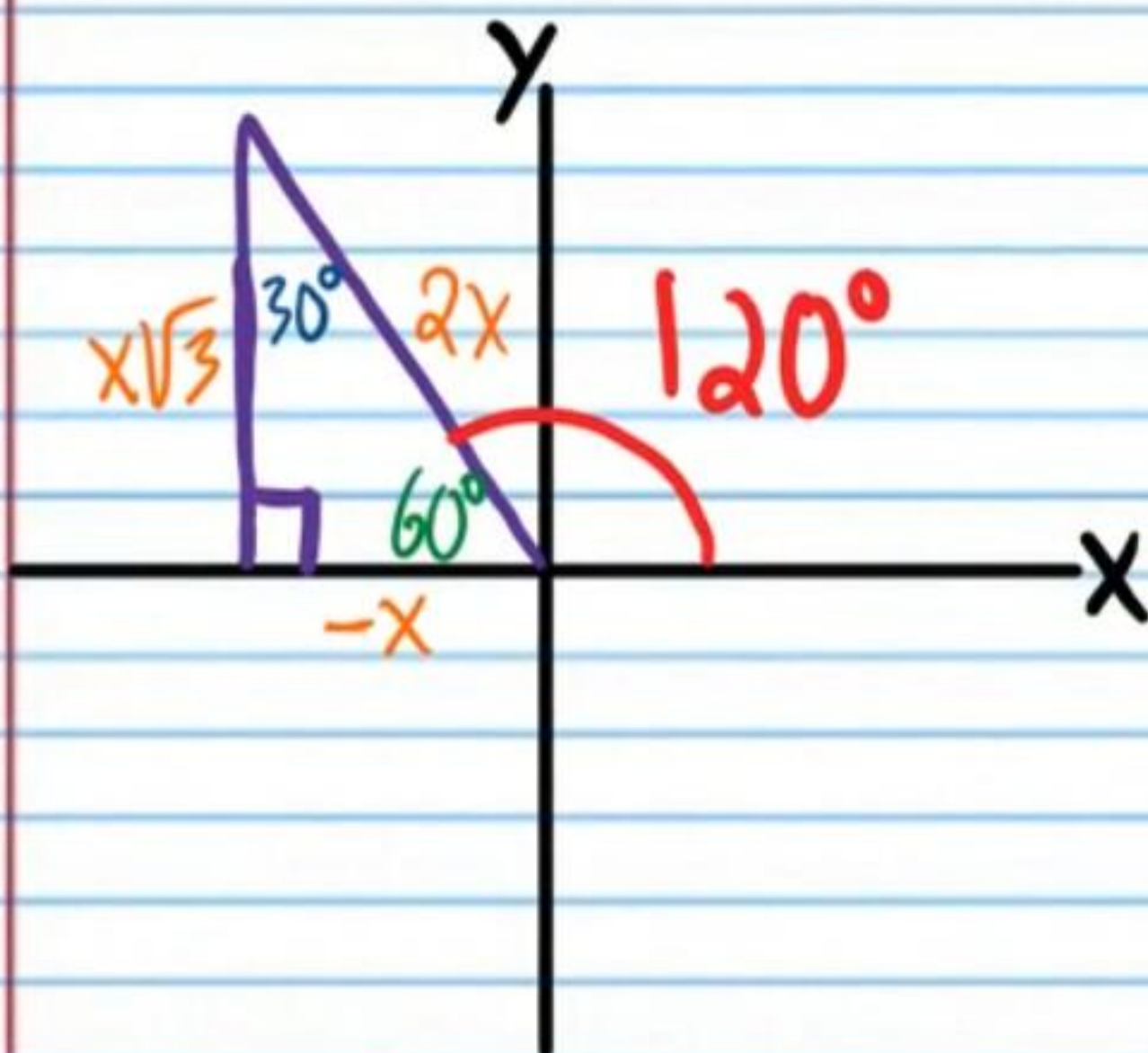
IV) You will need a calculator for this test (be sure that it is in degree mode).

V) Each problem is worth four points for a total of 100 points.

Find the values of the following expressions without a calculator. You must write exact answers.

Draw each angle in standard position. Write the measure of each angle and the measure of its reference angle, along with the corresponding right triangle. If a triangle is collapsed, then you cannot draw it. In this scenario, draw only the angle.

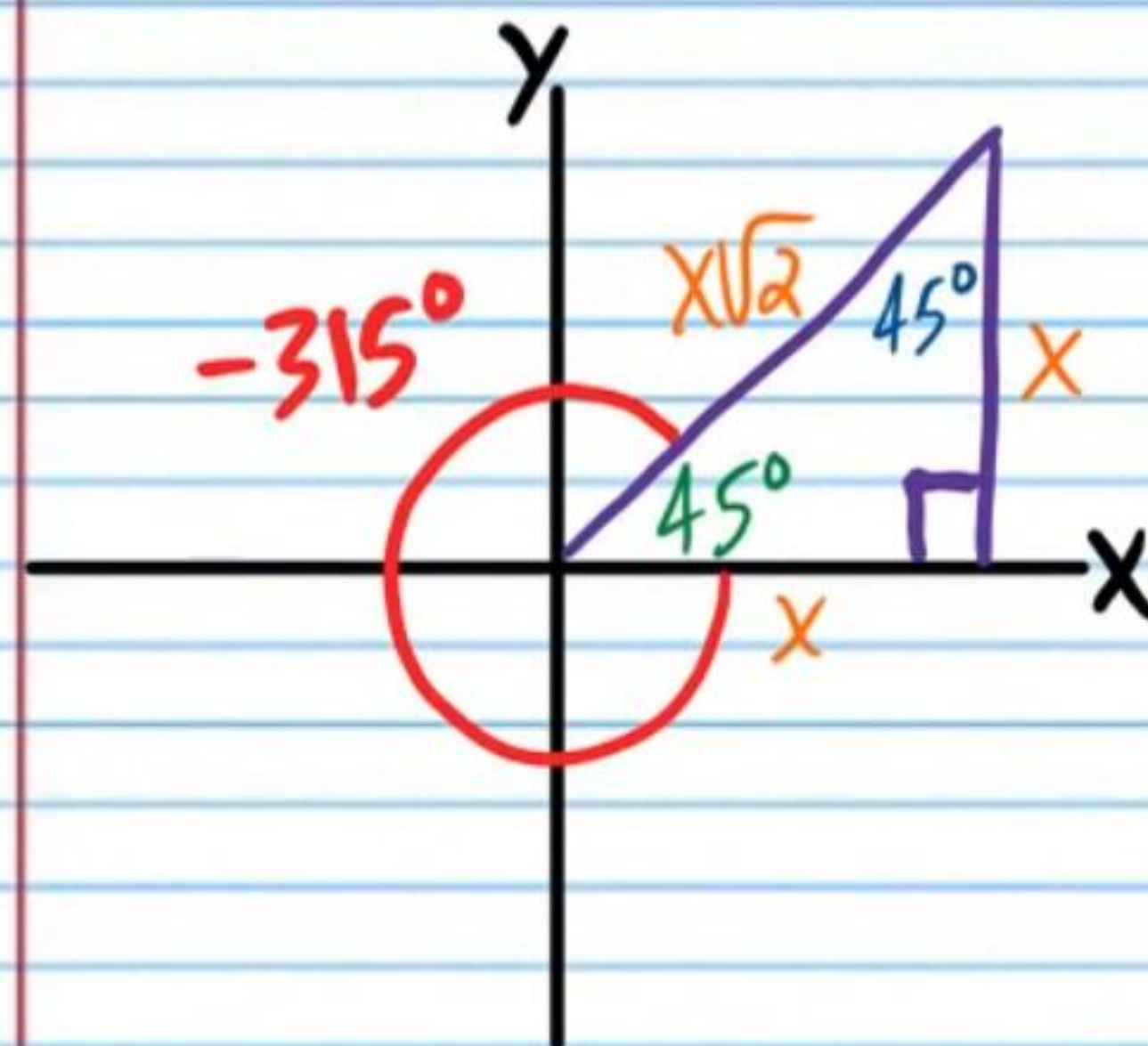
① $\sin 120^\circ$



$$\sin 120^\circ = \frac{x\sqrt{3}}{2x}$$

$$= \boxed{\frac{\sqrt{3}}{2}}$$

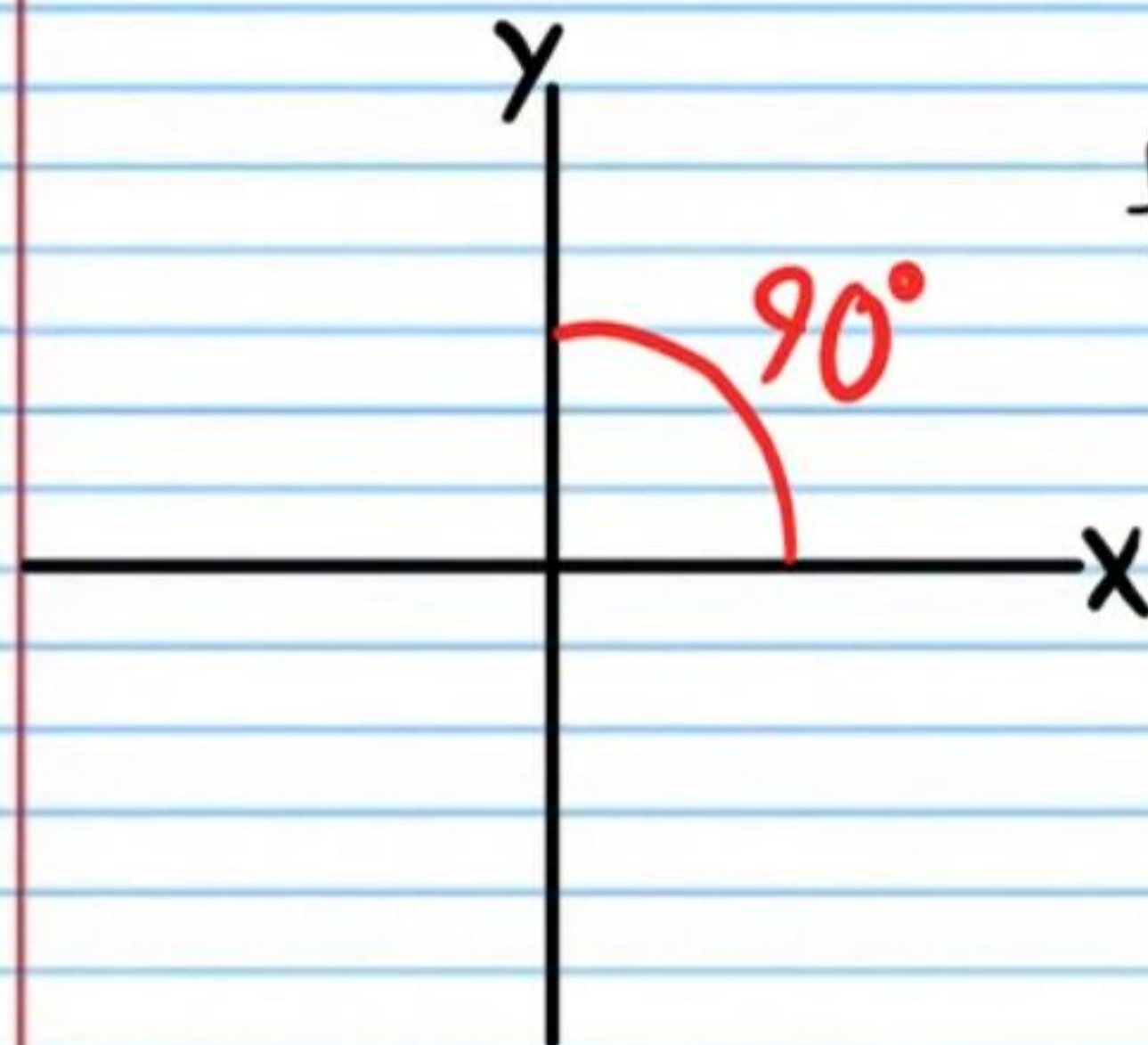
② $\sec(-315^\circ)$



$$\sec(-315^\circ) = \frac{x\sqrt{2}}{x}$$

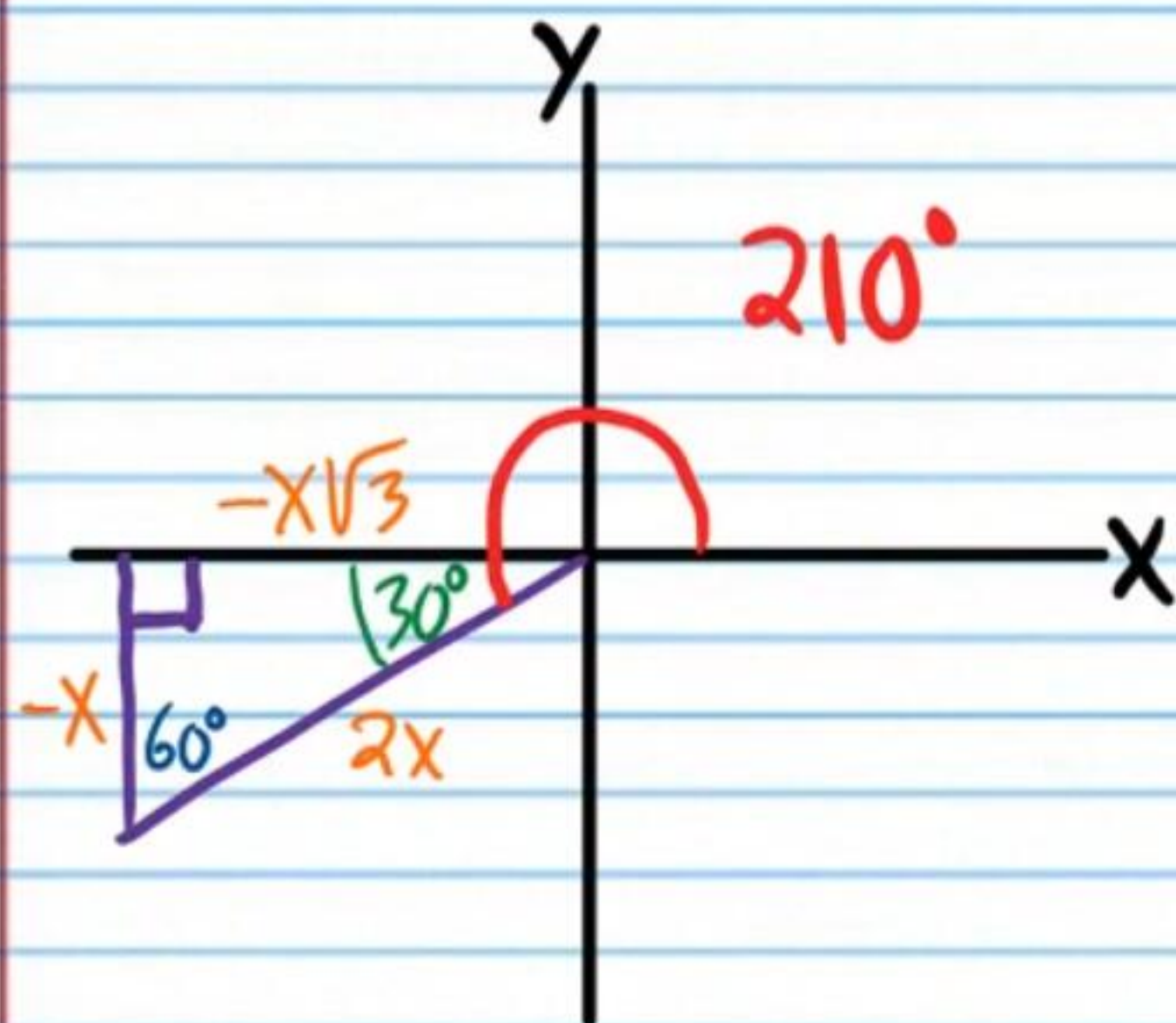
$$= \boxed{\sqrt{2}}$$

③ $\tan 90^\circ$



$$\tan 90^\circ = \boxed{\text{Undefined}}$$

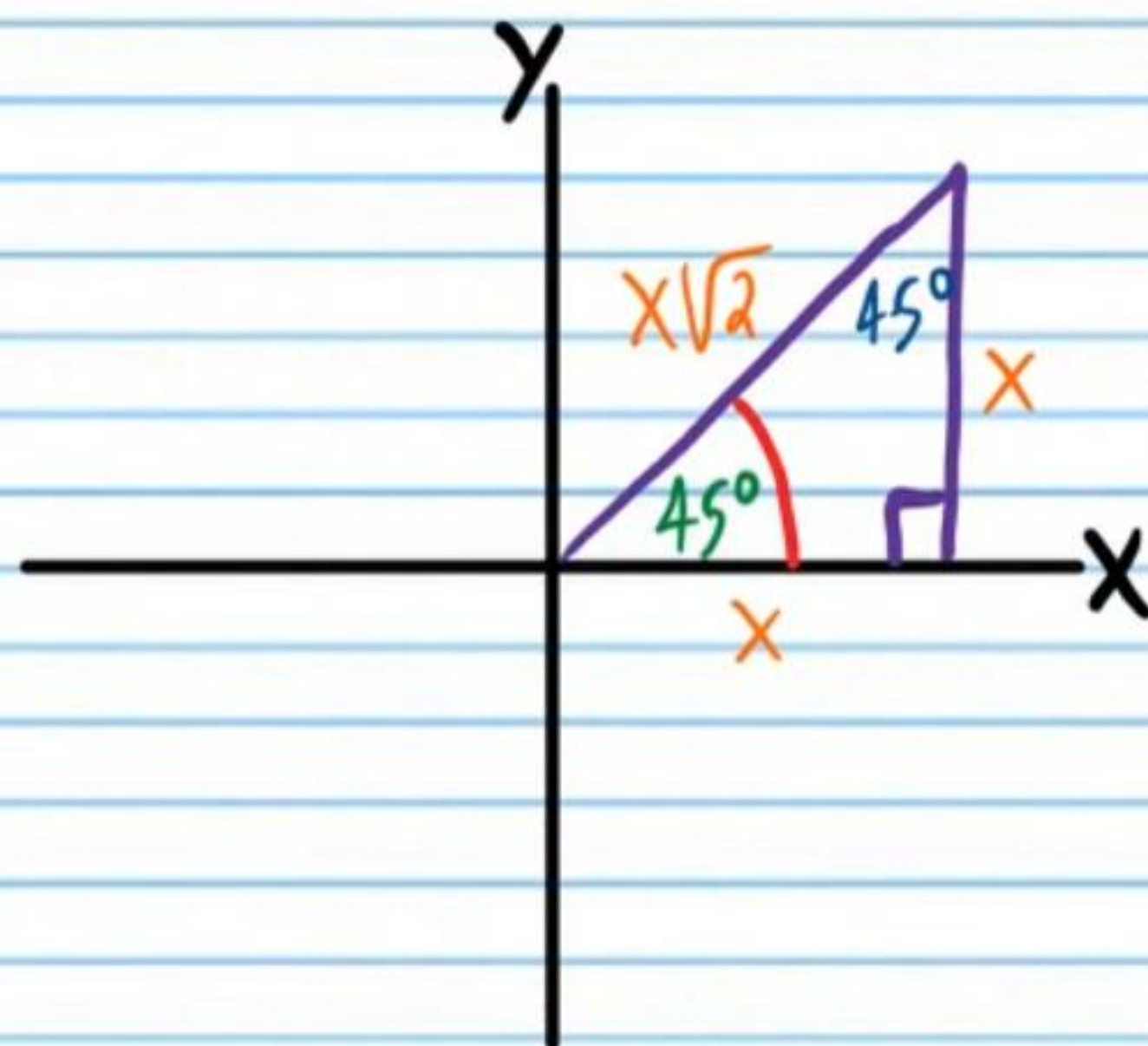
④ $\cot 210^\circ$



$$\cot 210^\circ = \frac{-x\sqrt{3}}{-x}$$

$$= \boxed{\sqrt{3}}$$

⑤ $\cos 45^\circ$

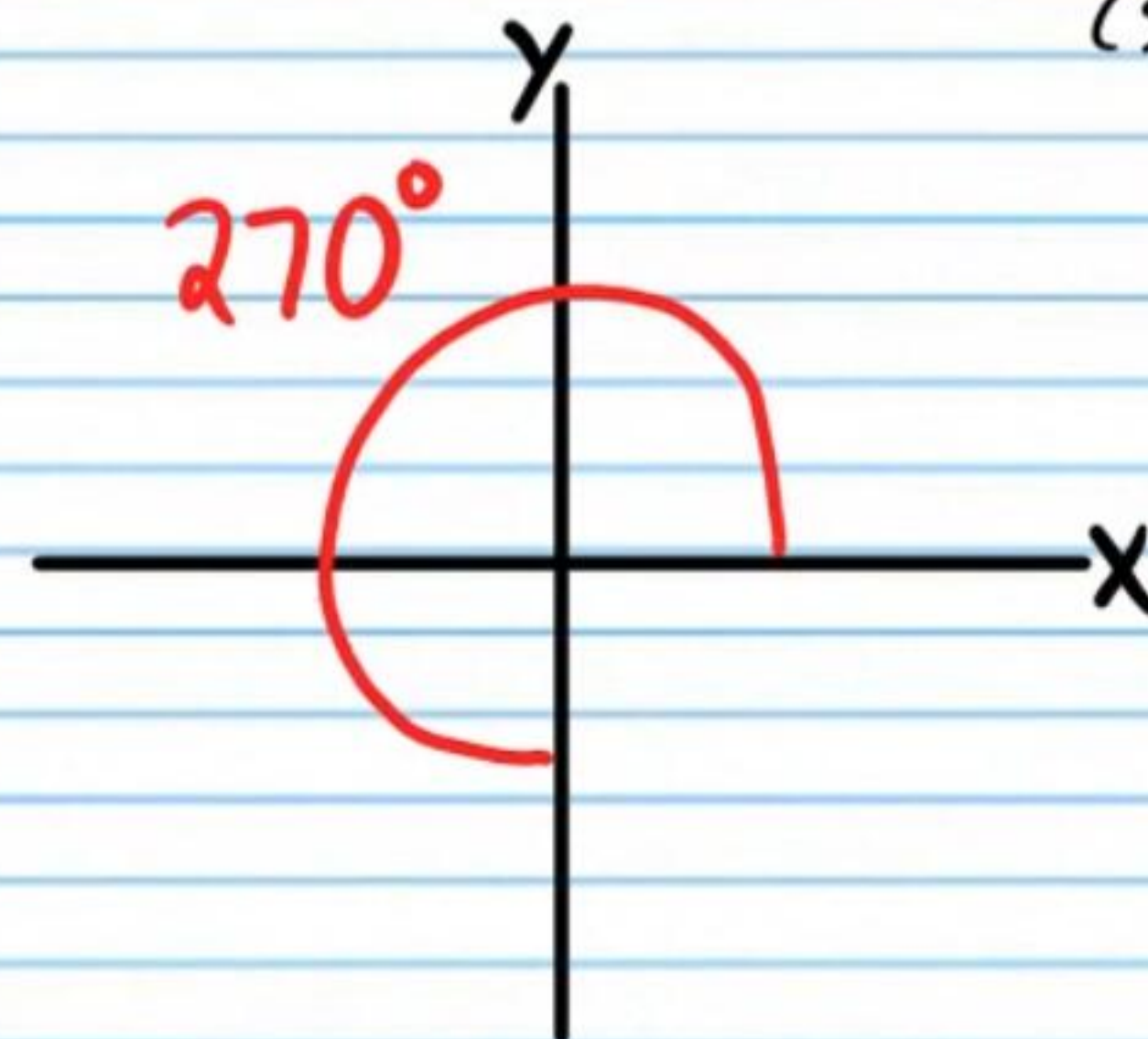


$$\cos 45^\circ = \frac{x}{x\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}$$

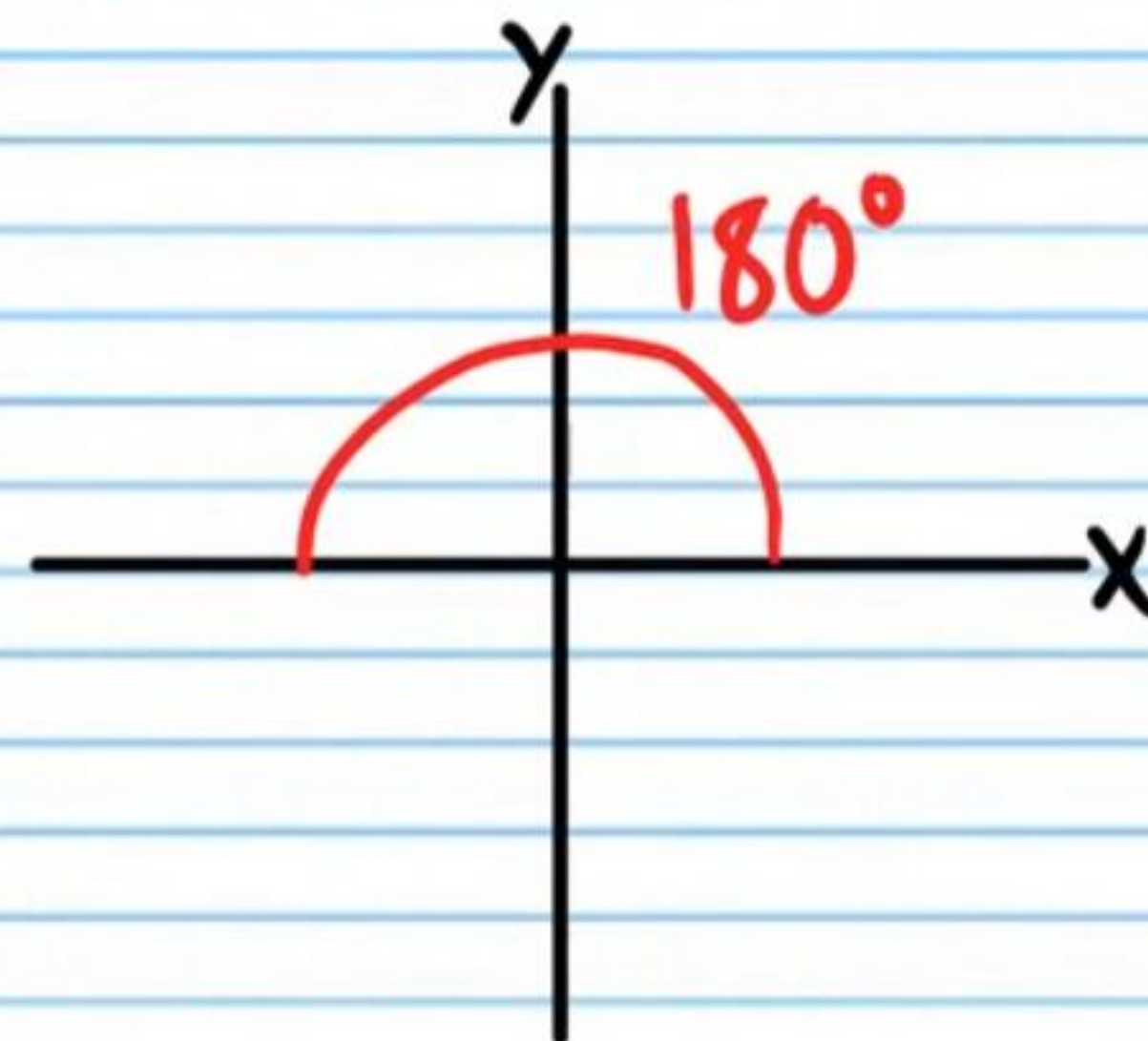
$$= \boxed{\frac{\sqrt{2}}{2}}$$

⑥ $\csc 270^\circ$



$$\csc 270^\circ = \boxed{-1}$$

⑦ $\sin 180^\circ$



$$\sin 180^\circ = \boxed{0}$$

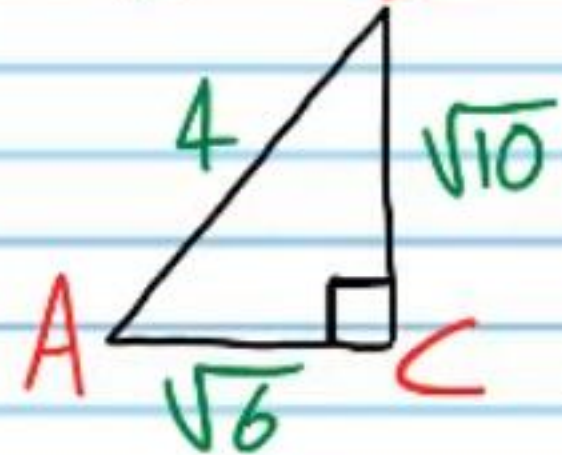
⑧ Find the values of the expressions below **with a calculator**. Round to the nearest thousandth.

i) $\sin 127^\circ \approx \boxed{0.799}$

ii) $\tan 35^\circ \approx \boxed{0.7}$

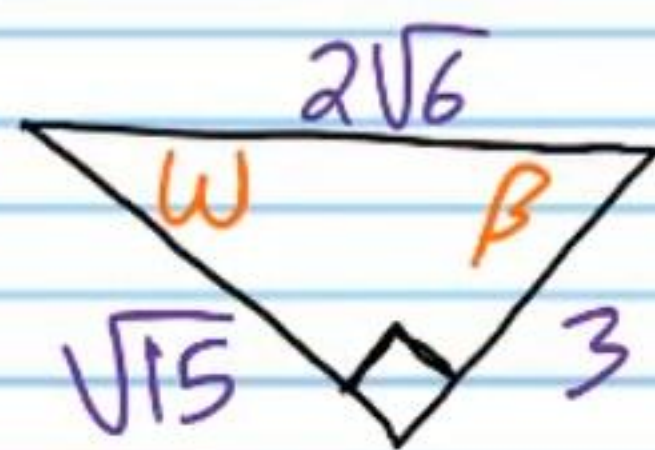
⑨ Use the triangles below to find the values of the expressions. **If necessary, rationalize denominators.**

i) $\cos A$



$$\cos A = \boxed{\frac{\sqrt{6}}{4}}$$

ii) $\cot \beta$



$$\cot \beta = \frac{3}{\sqrt{15}}$$

$$\cot \beta = \frac{3\sqrt{15}}{15}$$

$$\cot \beta = \boxed{\frac{\sqrt{15}}{5}}$$

⑩ The measure of an angle in standard position is written below. Find the measures of three coterminal angles in standard position.

$$244^\circ$$

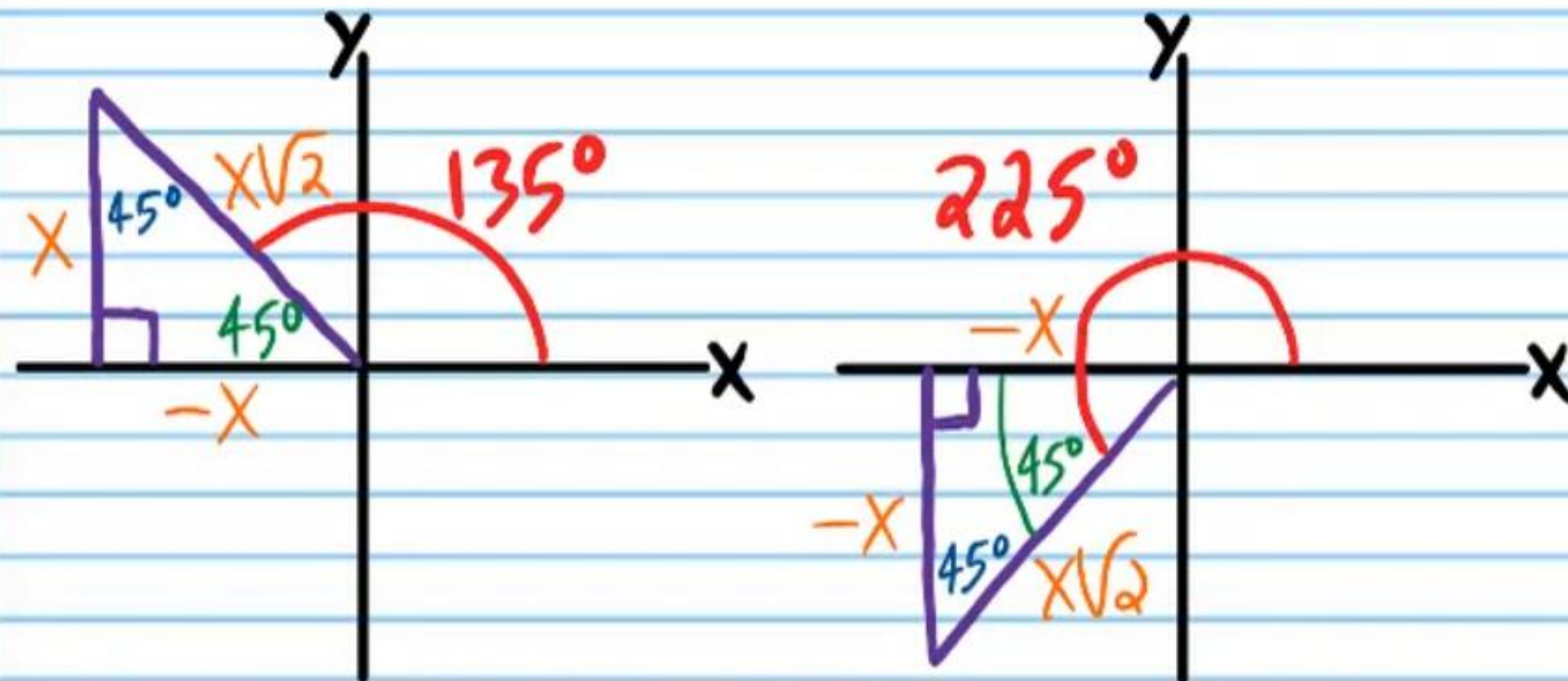
$$\theta_1 = 244^\circ + 360^\circ = \boxed{604^\circ}$$

$$\theta_2 = 244^\circ - 360^\circ = \boxed{-116^\circ}$$

$$\theta_3 = 244^\circ + 360^\circ + 360^\circ = \boxed{964^\circ}$$

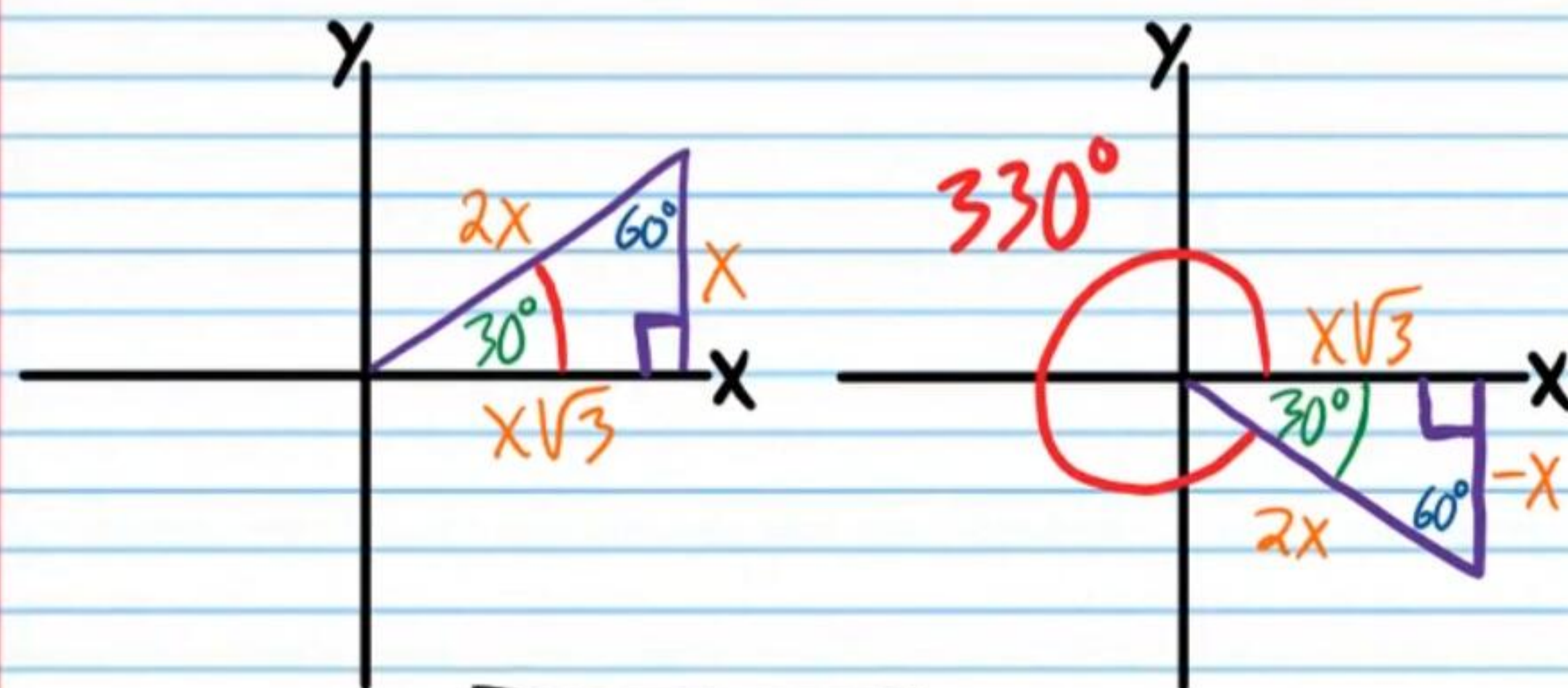
For problem #'s 11-15, draw the angles that are solutions to each equation in standard position as well as their corresponding right triangles. Write all three interior angle measures of each triangle and the relative side lengths. **Do not use a calculator.**

⑪ If $\sec \theta = -\sqrt{2}$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ .



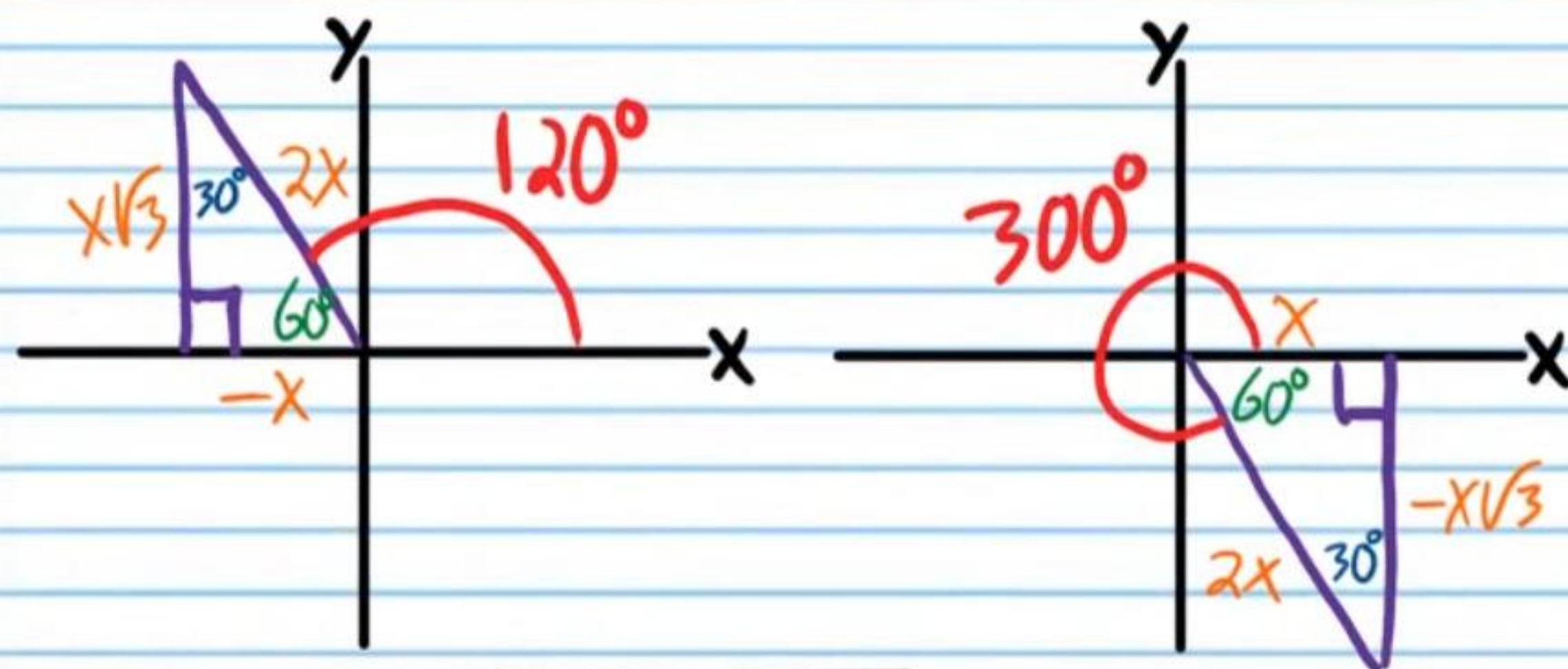
$$\theta = \boxed{135^\circ, 225^\circ}$$

⑫ If $\cos\theta = \frac{\sqrt{3}}{2}$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ .



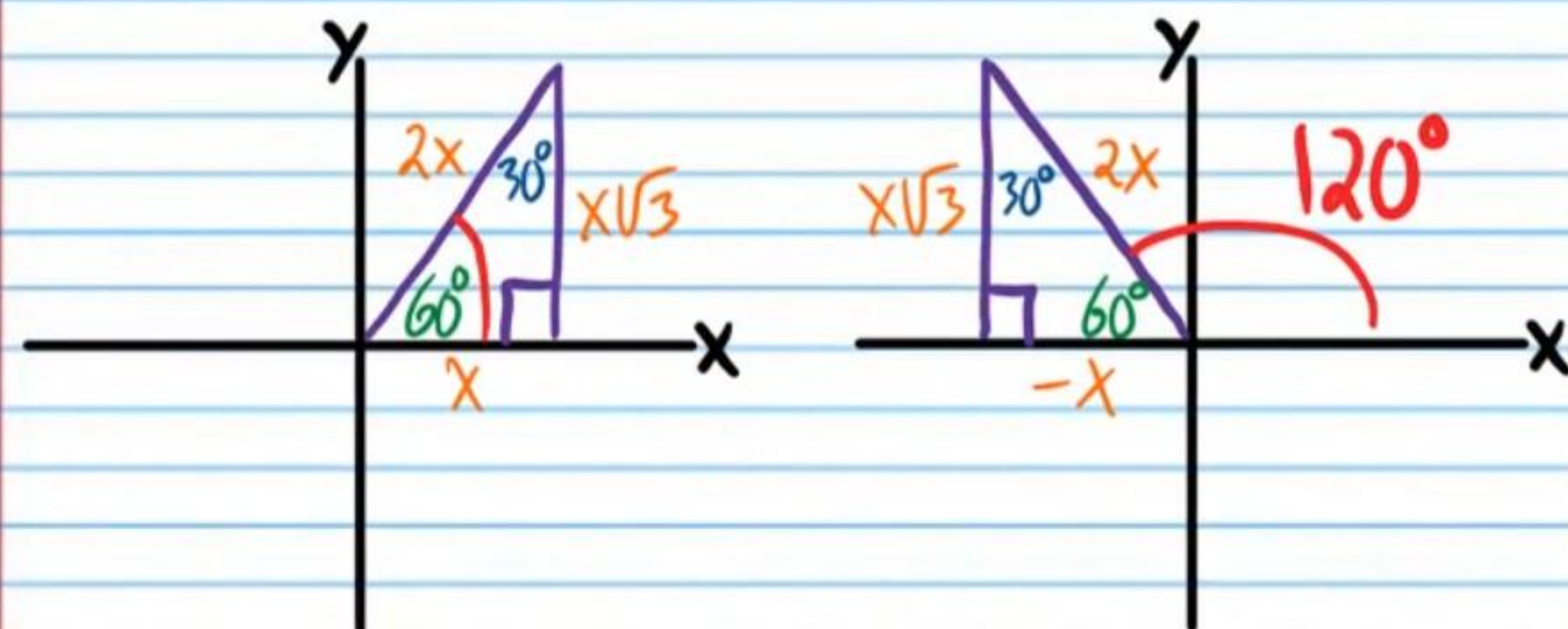
$$\theta = 30^\circ, 330^\circ$$

⑬ If $\tan\theta = -\sqrt{3}$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ .



$$\theta = 120^\circ, 300^\circ$$

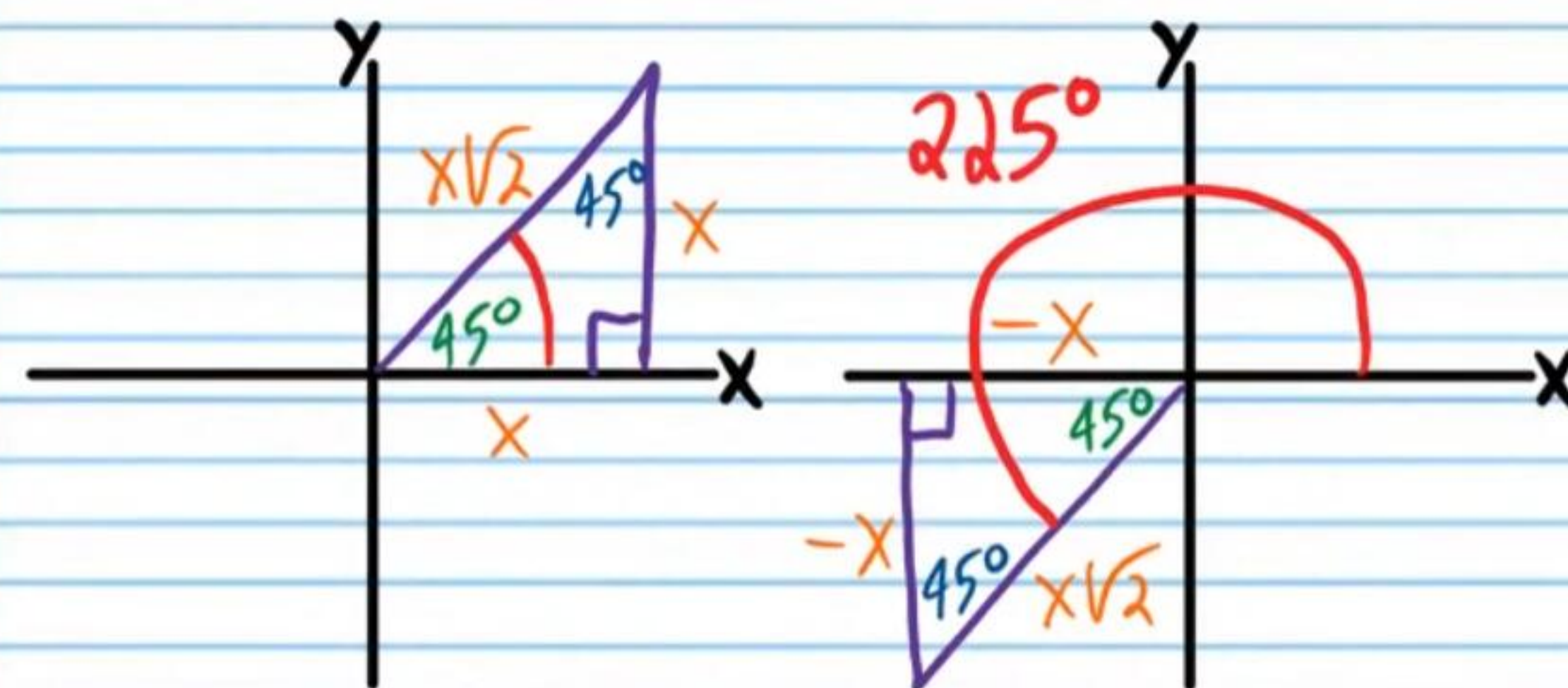
⑭ If $\csc\theta = \frac{2}{\sqrt{3}}$, find all values of θ .



$$\theta = 60^\circ + 360n, n \text{ is any integer}$$

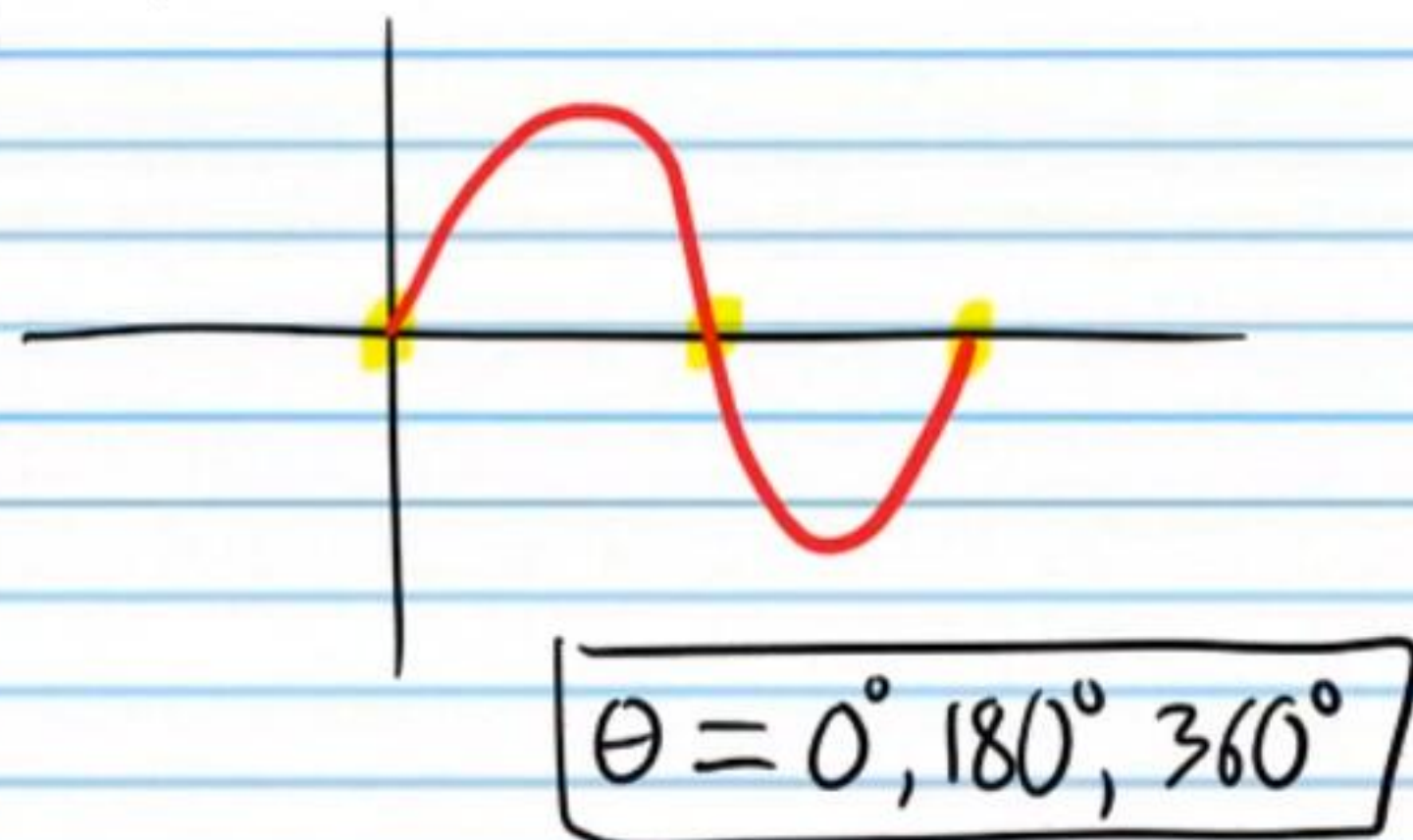
$$\theta = 120^\circ + 360n, n \text{ is any integer}$$

⑮ If $\cot\theta = 1$, find all values of θ .



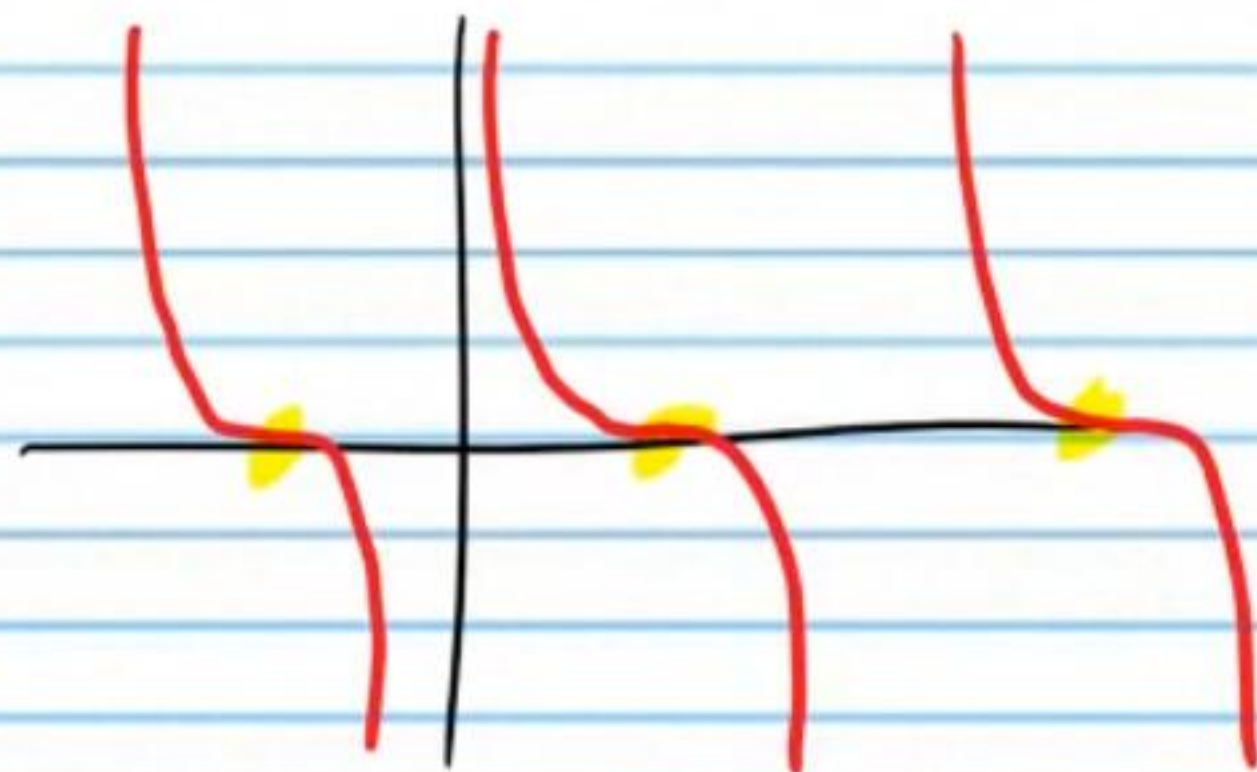
$$\theta = 45^\circ + 180n, n \text{ is any integer}$$

⑩ If $\sin \theta = 0$, and $0 \leq \theta \leq 360^\circ$, find the values of θ .



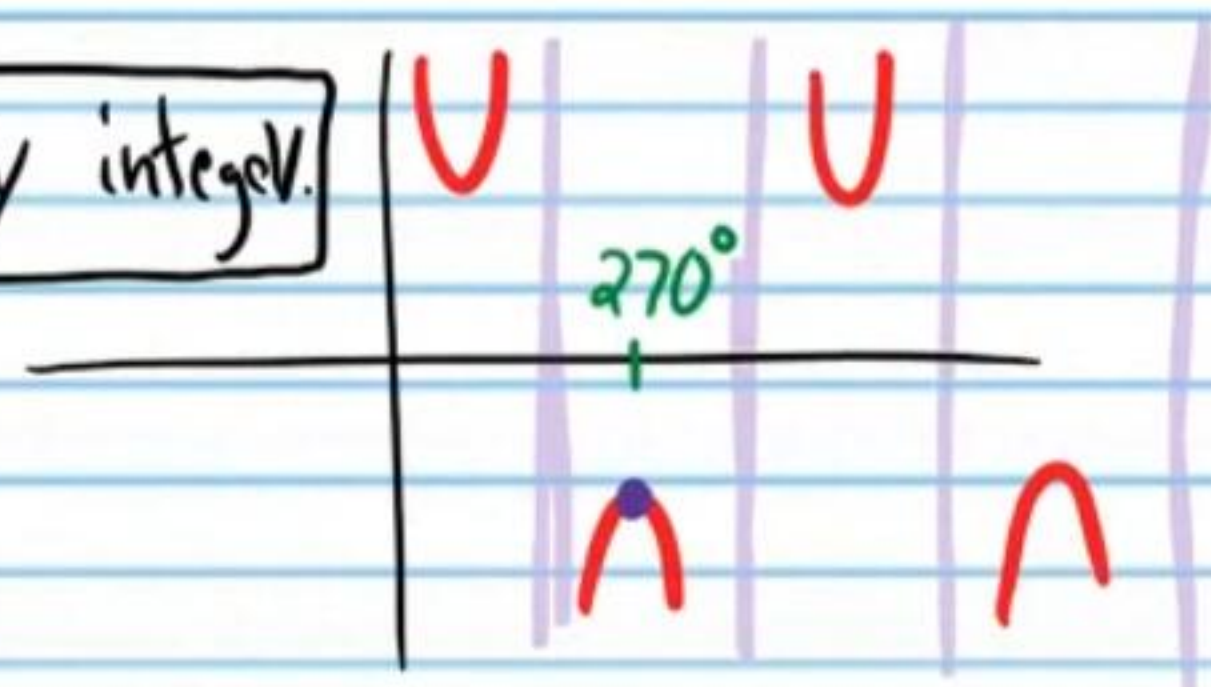
⑪ If $\cot \theta = 0$, and $0 \leq \theta \leq 360^\circ$, find the values of θ .

$\theta = 90^\circ, 270^\circ$



⑫ If $\csc \theta = -1$, find all values of θ .

$\theta = 270^\circ + 360n$, n is any integer.

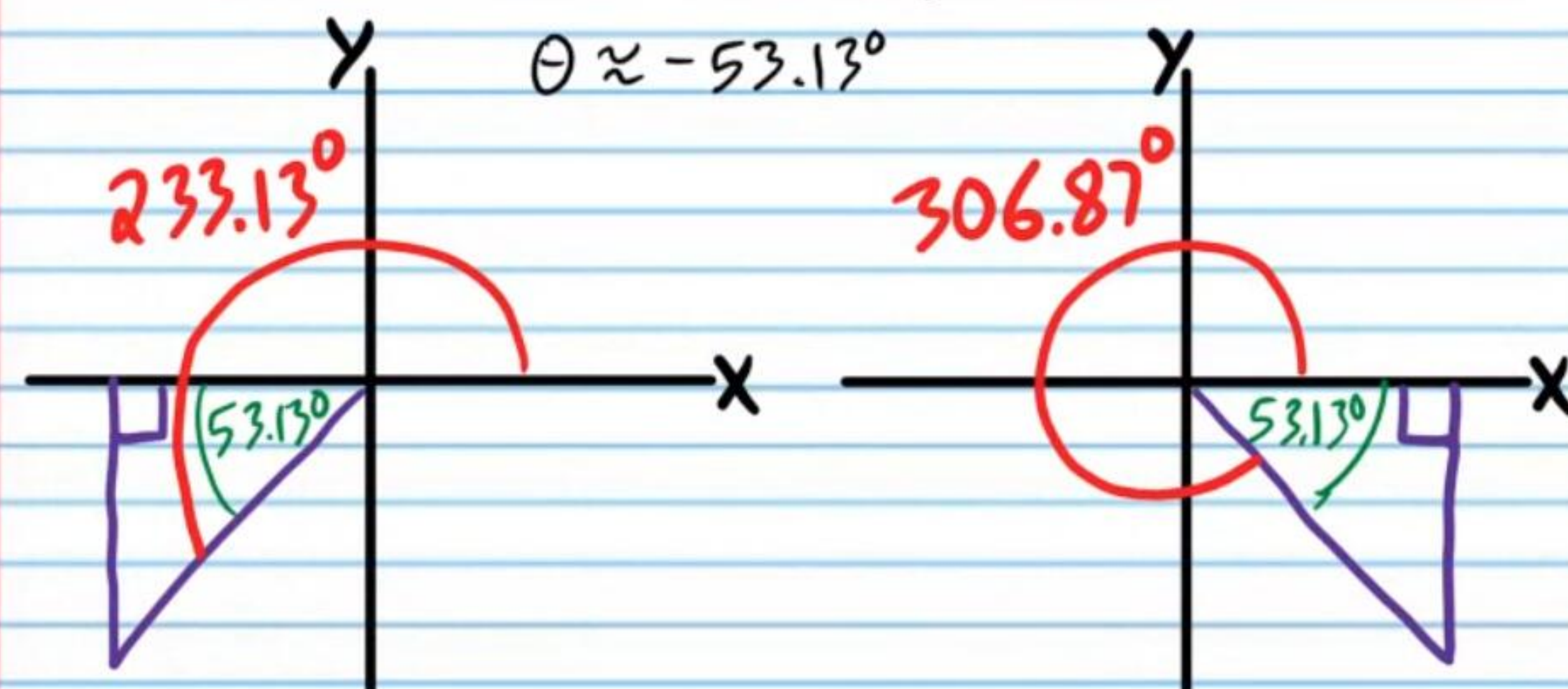


For problem #'s 19-21, use a calculator to find the degree measures. Draw each angle in standard position, the corresponding right triangle, and the reference angle.

⑲ If $\sin \theta = -\frac{4}{5}$, and $0^\circ < \theta \leq 360^\circ$, find the values of θ .

$\theta = \sin^{-1}\left(-\frac{4}{5}\right)$

$\theta \approx -53.13^\circ$

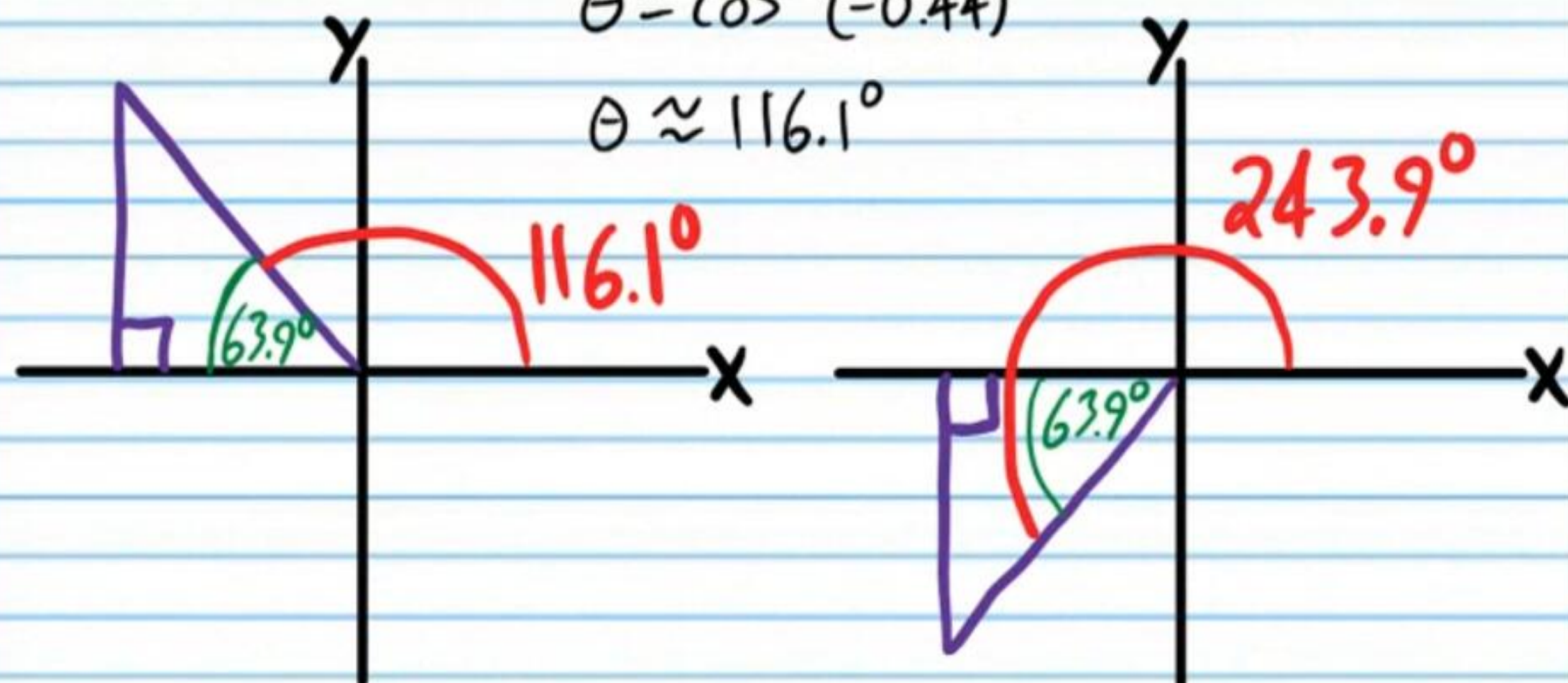


$\theta \approx 233.13^\circ, 306.87^\circ$

20) If $\cos \theta = -0.44$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ .

$$\theta = \cos^{-1}(-0.44)$$

$$\theta \approx 116.1^\circ$$

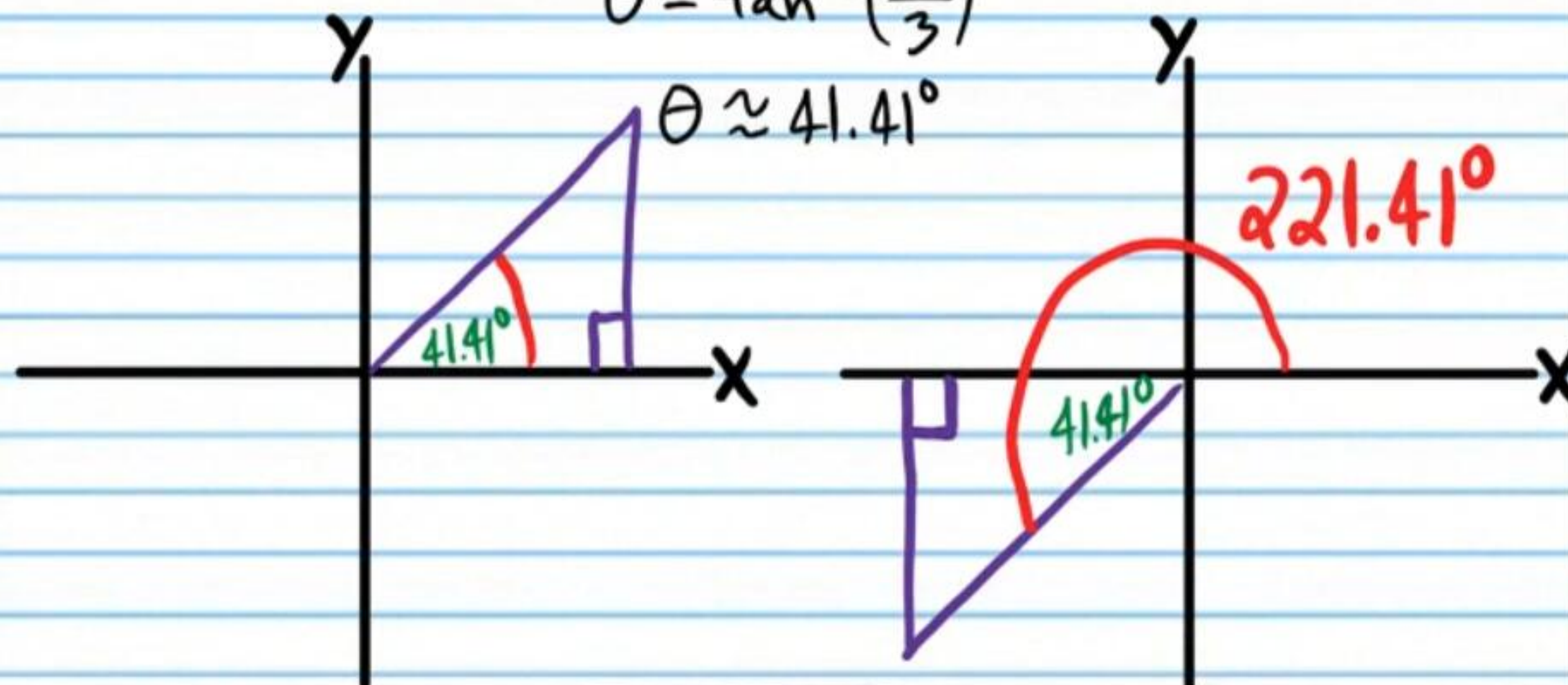


$$\theta \approx 116.1^\circ, 243.9^\circ$$

21) If $\tan \theta = \frac{\sqrt{7}}{3}$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ .

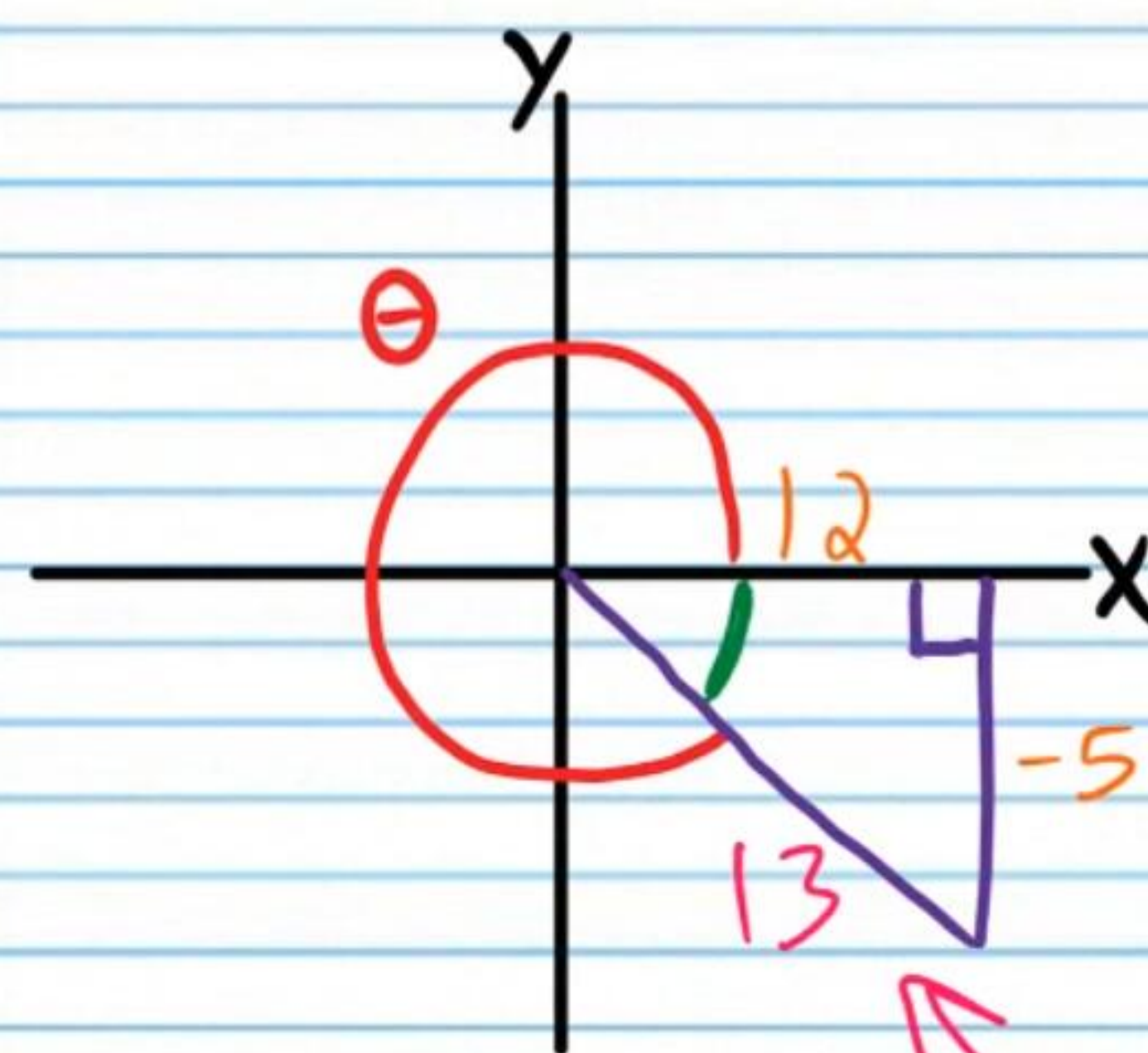
$$\theta = \tan^{-1}\left(\frac{\sqrt{7}}{3}\right)$$

$$\theta \approx 41.41^\circ$$



$$\theta \approx 41.41^\circ, 221.41^\circ$$

22) If $\cot \theta = -\frac{12}{5}$, and θ terminates in quadrant IV, find the values of the five remaining basic trigonometric functions of θ .



$$a^2 + b^2 = c^2$$

$$12^2 + (-5)^2 = c^2$$

$$144 + 25 = c^2$$

$$169 = c^2$$

$$\pm 13 = c$$

$$c = 13$$

i) $\csc \theta = \boxed{-\frac{13}{5}}$

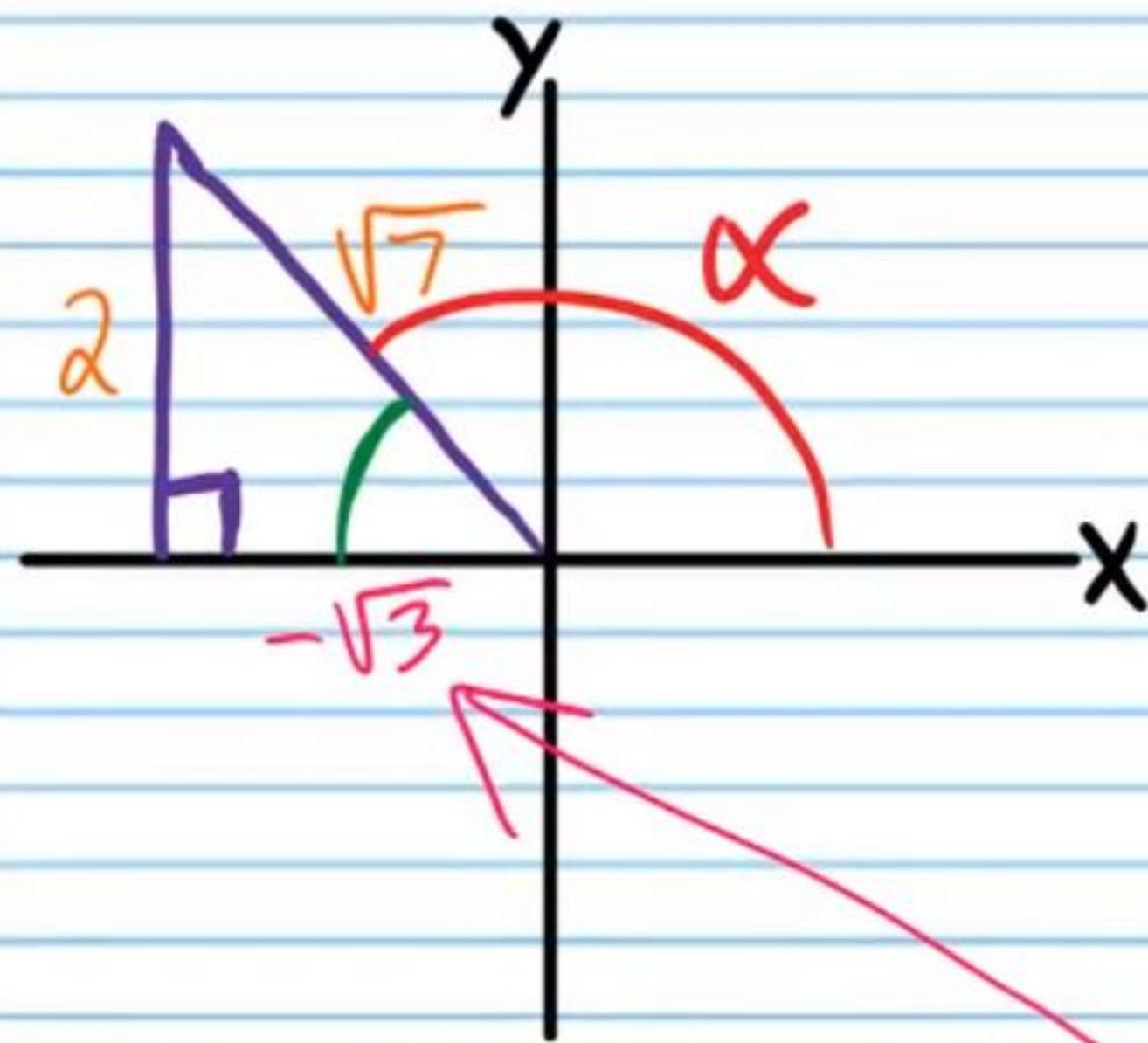
ii) $\sec \theta = \boxed{\frac{13}{12}}$

iii) $\sin \theta = \boxed{-\frac{5}{13}}$

iv) $\cos \theta = \boxed{\frac{12}{13}}$

v) $\tan \theta = \boxed{-\frac{5}{12}}$

23) If $\sin \alpha = \frac{2}{\sqrt{7}}$, and α terminates in quadrant II, find the values of the five remaining basic trigonometric functions of α .



$$a^2 + b^2 = c^2$$

$$2^2 + b^2 = (\sqrt{7})^2$$

$$4 + b^2 = 7$$

$$b^2 = 3$$

$$b = \pm \sqrt{3}$$

$$b = -\sqrt{3}$$

$$\text{i) } \csc \alpha = \boxed{\frac{\sqrt{7}}{2}}$$

$$\text{ii) } \sec \alpha = -\frac{\sqrt{7}}{\sqrt{3}} = \boxed{-\frac{\sqrt{21}}{3}}$$

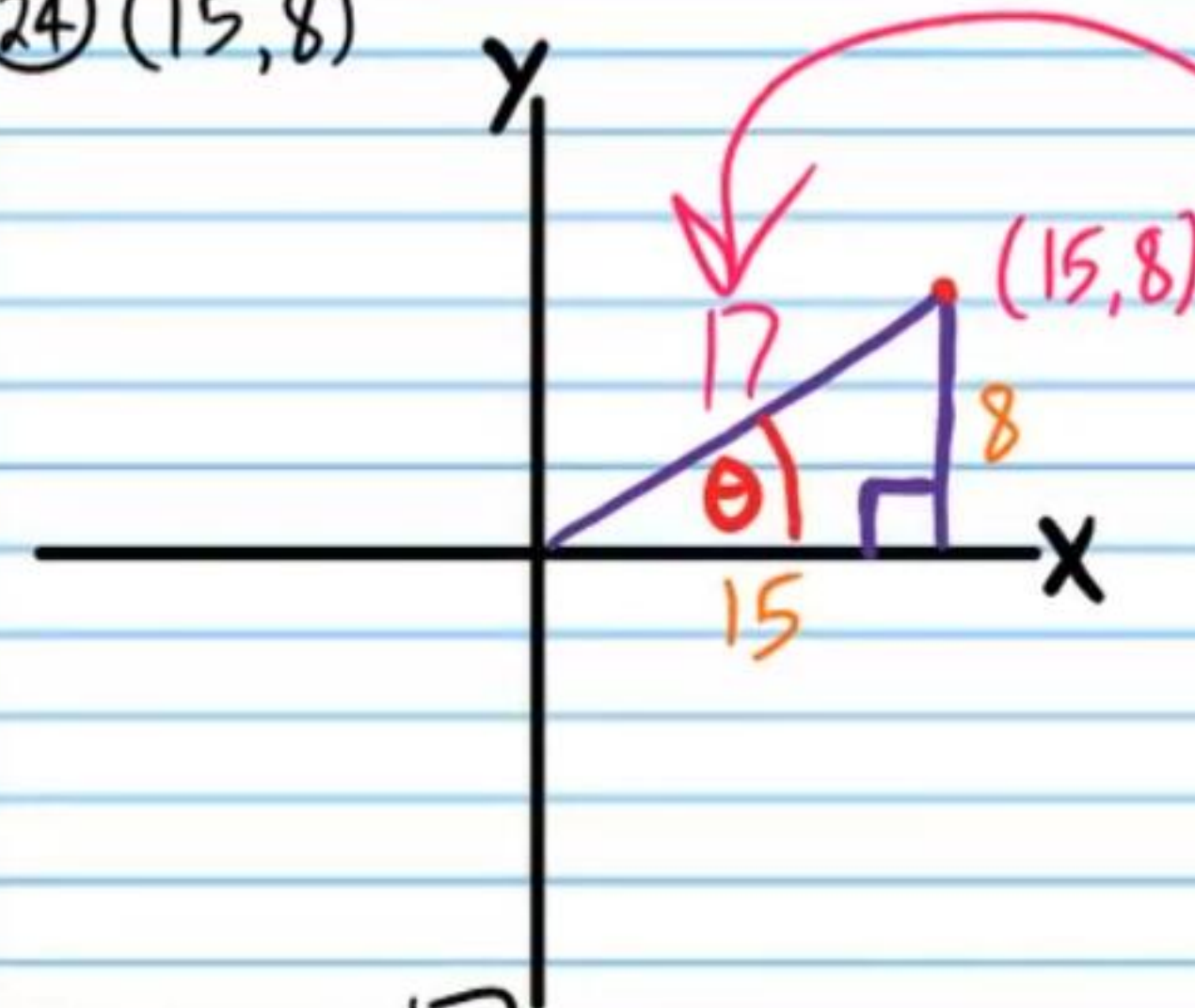
$$\text{iii) } \cot \alpha = \boxed{-\frac{\sqrt{3}}{2}}$$

$$\text{iv) } \cos \alpha = \frac{-\sqrt{3}}{\sqrt{7}} = \boxed{-\frac{\sqrt{21}}{7}}$$

$$\text{v) } \tan \alpha = \frac{2}{-\sqrt{3}} = \boxed{-\frac{2\sqrt{3}}{3}}$$

The coordinates of points are given below. If each of these points lies on the terminal side of an angle in standard position, find the values of the six basic trigonometric functions of the angle.

24) (15, 8)



$$a^2 + b^2 = c^2$$

$$15^2 + 8^2 = c^2$$

$$225 + 64 = c^2$$

$$289 = c^2$$

$$\pm 17 = c$$

$$17 = c$$

$$\text{i) } \sin \theta = \boxed{\frac{8}{17}}$$

$$\text{ii) } \cos \theta = \boxed{\frac{15}{17}}$$

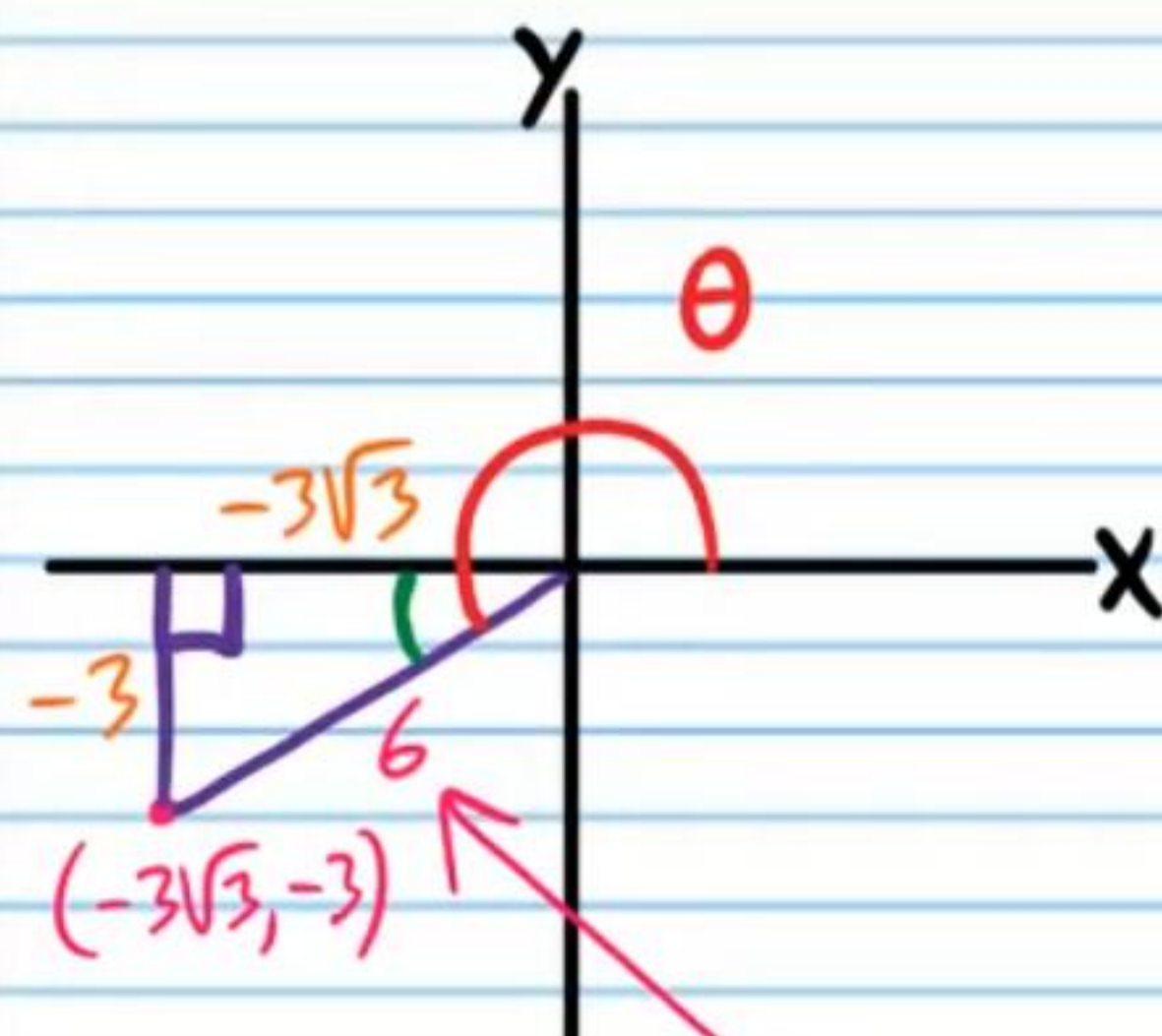
$$\text{iii) } \tan \theta = \boxed{\frac{8}{15}}$$

$$\text{iv) } \csc \theta = \boxed{\frac{17}{8}}$$

$$\text{v) } \sec \theta = \boxed{\frac{17}{15}}$$

$$\text{vi) } \cot \theta = \boxed{\frac{15}{8}}$$

Q5) $(-3\sqrt{3}, -3)$



$$a^2 + b^2 = c^2$$

$$(-3\sqrt{3})^2 + (-3)^2 = c^2$$

$$(-3\sqrt{3})(-3\sqrt{3}) + 9 = c^2$$

$$9\sqrt{9} + 9 = c^2$$

$$9(3) + 9 = c^2$$

$$27 + 9 = c^2$$

$$36 = c^2$$

$$\pm 6 = c$$

i) $\sin \theta = \frac{-3}{6} = \boxed{-\frac{1}{2}}$

ii) $\cos \theta = \frac{-3\sqrt{3}}{6} = \boxed{-\frac{\sqrt{3}}{2}}$

iii) $\tan \theta = \frac{-3}{-3\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$

iv) $\csc \theta = \frac{6}{-3} = \boxed{-2}$

v) $\sec \theta = \frac{6}{-3\sqrt{3}} = -\frac{2}{\sqrt{3}} = \boxed{-\frac{2\sqrt{3}}{3}}$

vi) $\cot \theta = \frac{-3\sqrt{3}}{-3} = \boxed{\sqrt{3}}$